Beyond Factual Accuracy: Detecting Misleading AI Answers Through Omission and Context[1]

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Abstract

LLM answers to administrative questions can read well yet name the wrong body, miss a deadline, or lean on an outdated rule. We test whether an evaluator with an institution-aware rubric (C) catches such problems better than a strong conventional fact-checker (B), when both share the same base model, reference pack of Brussels authorities and sources, and output format. It builds on our ICML 2026 study. We gathered 54 answers from two systems to three Brussels questions in English, French and Dutch (D1), and built 36 controlled objects with injected errors and benign edits (D2). Of the 54 D1 answers, 18 were gold-annotated blind (33 issues, 14 material), and flags were matched blind. On these 18, B and C each raised 61 flags, 56% judged valid (gold or newly found), and each found 20 of 33 gold issues. They found 6 (B) and 7 (C) of 14 material ones. At answer level C reached recall 1.00 and precision 0.67 (B: 0.88, 0.58), and its severity agreement was higher (0.65 vs 0.44), with overlapping intervals. On D2 both caught all six factual and all six authority perturbations, 5 of 6 source and 3 of 6 omission ones, and flagged none of the six benign edits. The gold sample showed no material action-epistemic gap: all eight materially problematic answers failed the practical-adequacy checklist. Both struggled with controlled omissions, and reliance on a frozen reference pack limits detection outside its coverage. Samples are small; results are descriptive.

1. Introduction

A person asking an AI assistant how to register as a jobseeker in Brussels, or whether an integration course is mandatory, needs more than correct sentences. They need the right body (a regional office, not a federal one), the right level of government, a deadline anchored to the right event, and nothing missing that would make them skip a step. A check that tests individual claims against general web evidence can pass an answer that names the wrong authority, relies on a superseded source, or silently omits a step. Our earlier work on interpretive fidelity argued that this gap exists and measured it by hand on 18 chat answers (Section 3.2). This project asks a narrower, testable question: when a strong claim-by-claim evaluator is given the same institutional reference, does an evaluator built around institutional fidelity still find problems that it misses?

The comparison is built to isolate the method. Evaluator B decomposes claims, verifies them, checks attribution and coverage. Evaluator C applies an institutional rubric: authority and operator, jurisdiction and level, deadlines and exemptions, source fit, a necessity test for omissions, and framing. Both run on the same base model and parameters, receive the same reference pack (authority map, answer keys, sources registry), see answers without system labels, and return the same JSON schema. Any difference therefore comes from the procedure, not from access to better information.

Our main contributions are:

1. A controlled B-versus-C comparison with a frozen, tagged protocol, in which the baseline receives the same institutional reference as the specialised method.

1 Research conducted at the AI x Epistemics Research Sprint, November 2026

2. **Three datasets:** D1 (54 web-search answers, 3 questions × EN/FR/NL × 2 systems × 3 runs) with a blind gold annotation of 18 answers (33 issues); D2 (36 controlled objects with injected errors); and the D0 development set from prior work.

3. **A negative but clean result:** on these data the institutional rubric did not detect more issues than the baseline. Its possible edge is severity calibration (0.65 vs 0.44 agreement), and both evaluators share the same blind spots (omissions, rules absent from the key).

Prior work and sprint work. Pre-existing: the ICML 2026 Culture × AI study, its 18 chat responses (D0), the interpretive-fidelity rubric and its coding instrument, and the Q1 prompt. Created during the sprint (13–15 Nov 2026): prompts Q2–Q3 and the reference key, D1, the blind gold, D2, evaluators B and C, all code in the repository, the blind matching and every result reported here. Details in Appendix B.

2. Related Work

Factuality evaluators decompose a long answer into atomic claims and verify each against evidence, as in FActScore [1] and the search-augmented SAFE method [2]. Work on attribution and citation asks whether a cited source supports the statement it is attached to [3, 4, 5]. LLM judges are widely used as scalable evaluators, with known agreement and bias issues [6]. These approaches treat correctness as a property of claims and their citations.

Our target is different: whether the answer places responsibility with the right institution, at the right level and territory, under the current rule, and whether what it leaves out would change the user's next step. This is the interpretive fidelity construct of our earlier study, which combines factual accuracy, source fidelity, contextual meaning and cross-linguistic consistency. We do not claim a new benchmark for general factuality. Our contribution is an experimental test of one specific design choice, institution-aware rubric versus a strong decomposition baseline with identical reference material, in a domain (Belgian and Brussels administration, three languages) where authority and level are the main failure sources.

3. Methods

3.1 Design and freezes. The protocol (v2.1), prompts, reference key, evaluators, D1 collection, gold and D2 were each frozen with a git tag before the next stage (protocol-v2.1, reference-v1.0, prompts-v1.0, evaluators-v1.0, d1-collected, gold-v1.0, d2-v1.0). These are repository tags, not an external registry. Evaluator instructions were tuned only on D0 (18 answers, one run) and left unchanged; deviations are logged in CHANGELOG.md.

3.2 Data. D0 is the 18 ICML chat responses (Q1, GPT and Claude consumer interfaces, EN/FR/NL × 3 runs) and serves only as development set. D1 uses three questions about a synthetic user profile in Brussels: Q1 mandatory integration programme, Q2 jobseeker registration and benefits after dismissal, Q3 recognition of a foreign Master's degree. Each is asked in EN, FR and NL (equivalence checked by an independent reviewer). Systems: OpenAI gpt-5.5 (reasoning effort low) and Anthropic claude-sonnet-5 (effort medium), both with hosted web search and Brussels user location, 3 runs each, 54 answers, call order randomised. D2 takes six D1 answers outside the gold sample (two per question, three GPT, three Claude) and builds 36 objects: the original plus T1 factual error, T2 wrong authority, T3 wrong-jurisdiction or superseded source, T4 omission (deletions only) and T5 benign edit. Expected detector and severity were fixed on cards before any evaluator saw D2.

Table 1. Datasets. Gold answers were drawn before any evaluator saw D1: one run per question × language × system cell, seed 20261115.

Set	Content	n	Use
D0	ICML chat answers (pre-existing)	18	Evaluator tuning only
D1	API answers with web search	54	Main comparison (gold on 18)
Gold	Blind issue list on 18 D1 answers	33 issues	Reference for D1 metrics
D2	Controlled perturbations of 6 bases	36	Detection by error type

3.3 Evaluators. B and C run on claude-sonnet-5 (effort medium, no web search). Both receive the profile, authority map, the answer key and sources registry for the question under test; Q1 claims harvested from D0 and review notes were excluded. A shared file defines evidence rules, one omission rule (a missing checklist point is not material by itself; it is material only if the user could not take the correct next step), one source rule, a severity scale (minor, moderate, material) with five generic calibration examples, three coarse types (factual, source, omission_or_framing) and the output schema. B adds four steps: decompose, verify, attribute, check coverage. C adds six checks C1–C6 (authority, jurisdiction, deadlines/exemptions/language, sources, omission necessity, framing). Fine-grained error codes are optional and excluded from all comparisons.

3.4 Gold and matching. The author annotated the 18 gold answers blind to system, using the reference key, and listed 33 issues with a severity and a coarse type (mapped from free labels). Evaluator flags (122 on D1) were shuffled, stripped of evaluator, severity and type, and matched blind to gold issues, to NEW (a real problem not in gold) or to FA (false alarm). For D2, 230 flags were judged for whether they address the injected perturbation. An automatic first pass agreed with the human matching on 58 of 60 evaluator × object cells; the human matching is authoritative.

3.5 Metrics. Descriptive only, with Wilson 95% intervals and no significance tests: share of flags that are real problems, gold issues found, material issues found, material flags that are false alarms, severity agreement on matched pairs, and a response-level confusion matrix (does the evaluator flag a material problem versus gold). Cost of all evaluator runs was about $14 by script estimate (B $7.2, C $7.0).

4. Results

D1, issue level. Of the 54 D1 answers, 18 were gold-annotated (33 gold issues, 14 material); the numbers below refer to these 18. B and C raised the same number of flags (61 each); 34 of 61 each (56%) were judged valid, counting both matches with gold (B 24, C 23) and newly identified real problems (NEW). Each found 20 of 33 distinct gold issues; 18 are shared, two only by B (GI04, GI22), two only by C (GI14, GI28). Eleven gold issues were found by neither. No material flag was a false alarm for either evaluator. False alarms are mostly minor source flags (B 19 of 27, C 18 of 27). C is ahead on severity agreement, with overlapping intervals (Table 2, Figure 1).

Table 2. D1 issue-level results against the blind gold (18 answers, 33 issues, 14 material). Gold issues found = distinct issues. Brackets: Wilson 95% interval.

Metric (D1)	B	C
Flags raised	61	61
Flags judged valid (gold match or NEW)	34/61 = 0.56 [0.43, 0.67]	34/61 = 0.56 [0.43, 0.67]
Gold issues found	20/33 = 0.61 [0.44, 0.75]	20/33 = 0.61 [0.44, 0.75]
Material gold issues found	6/14 = 0.43 [0.21, 0.67]	7/14 = 0.50 [0.27, 0.73]
Material flags that are false alarms	0/20	0/18
Severity agreement on matched pairs	11/25 = 0.44 [0.27, 0.63]	15/23 = 0.65 [0.45, 0.81]

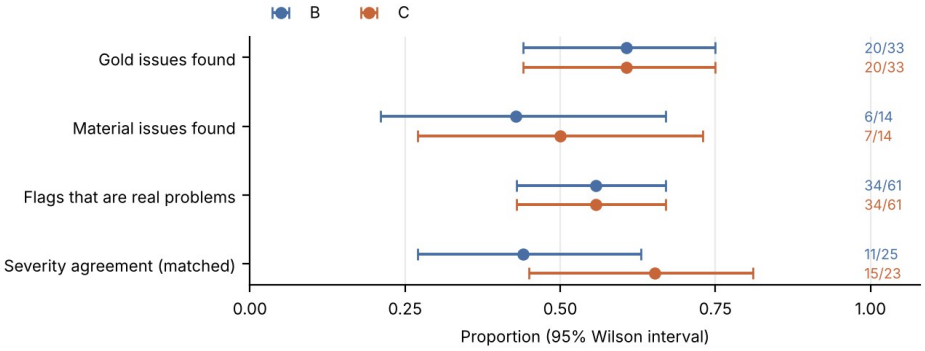


Figure 1. Main D1 metrics for evaluator B (blue) and C (orange) with Wilson 95% intervals. Intervals overlap on every metric; labels show counts.

By error type. Recall on gold issues by coarse type is almost identical (Table 3). The hypothesis that C would outperform B on omissions was not supported: both found 4 of 8 omission or framing issues, with one found only by each.

Table 3. D1 recall by gold coarse type.

Gold type	n	B found	C found	B only	C only
factual	23	14	15	0	1
source	2	2	1	1	0
omission_or_framing	8	4	4	1	1

D1, answer level. Eight of the 18 gold answers have a material problem. B flagged 7 of 8 with 5 false positives among the 10 clean answers (precision 0.58, recall 0.88); C flagged all 8 with 4 false positives (0.67, 1.00). The difference is one answer per cell. Across all 54 answers both evaluators flag Claude answers more often than GPT answers (B 21/27 vs 13/27, C 22/27 vs 12/27).

D2. Detection is the same for B and C (Table 4). Both miss the same four objects: Q1-NL-gpt T4, Q2-FR-gpt T4, Q3-EN-gpt T4 (omissions) and Q2-NL-claude T3 (source). Neither evaluator flagged any of the six benign edits (this concerns the edits only, not false alarms elsewhere in the original answers). Detected perturbations received the expected severity in 18 of 20 cases for B and 19 of 20 for C.

Table 4. D2 detection by perturbation type after blind human matching.

Type	Perturbation	B	C	Expected
T1	Factual error	6/6	6/6	B and C
T2	Wrong authority	6/6	6/6	B and C
T3	Wrong-jurisdiction / superseded source	5/6	5/6	B and C
T4	Omission (deletion)	3/6	3/6	mainly C
T5	Benign edit (should not flag)	0/6	0/6	none

Robustness. We report no significance tests. With 33 gold issues, 14 material ones and 6 objects per type, a one-item change moves a proportion by 3 to 17 points, so the gaps in Tables 2 to 4 are within noise. The matching procedure is stable (58 of 60 cells agree with the automatic pass). We did not vary prompts, models or seeds.

5. Discussion and Limitations

The result fits the argument that institutional errors are detectable, but it does not support the claim that a specialised rubric is needed for it. When the baseline sees the same authority map and answer key, it finds the same wrong authorities, wrong levels and outdated sources. This is consistent with the value lying in the institutional reference rather than in the procedure, but we did not run a baseline without the reference pack, so we cannot show it. The one plausible edge of C is severity calibration (0.65 vs 0.44), which matters because a user acts on severity, but the intervals overlap.

The shared blind spots matter more than the gap between B and C. Both miss half of the injected omissions and 11 of 33 gold issues, mostly subtle rules (for example, the effect of training on degressivity, or conditions on equivalence routes). These may simply be absent from the key, and evaluators are told not to judge outside it, so the ceiling is the key, not the model. Both also over-flag clean answers, mainly with minor source flags, though no material flag was false. For safety-relevant use, this suggests that evaluators of this kind are reasonable triage tools for material errors but cannot certify an answer as complete. In this sample the gold showed no action versus epistemic gap: every answer with a material problem was also practically incomplete or wrong, and none of the nine practically correct answers had one. This differs from our earlier study's conclusion and rests on 18 answers.

Limitations

Full list in Appendix A. One domain and three questions; 18 gold answers with 33 issues; one annotator who also wrote the reference key, so gold and evaluator pack share an origin; D2 built from six bases with perturbations written by the author; evaluators and one answer generator belong to the same model family; C differs from B in several ways at once (rubric and specialisation), so the test is of the method as a whole; and no significance claims are made.

Future Work

Add a no-reference baseline (B0) to separate the value of institutional knowledge from the rubric; independent second annotators; more questions and jurisdictions; an evaluator allowed to propose issues outside the key; and a test on consumer-interface answers as in D0.

6. Conclusion

In a frozen, blind comparison on Brussels administrative answers, an institution-aware evaluator and a strong conventional evaluator with the same reference material found the same share of real problems (56%) and the same share of gold issues (61%). C was slightly better on severity agreement and answer-level recall, but the observed differences were small and uncertain, with overlapping confidence intervals. Both evaluators struggled with controlled omissions, and their dependence on the frozen reference pack limits detection outside its coverage. The practical lesson for evaluation design is that the institutional reference, its coverage and the severity calibration deserve more effort than the evaluator procedure.

Code and Data

Code, prompts, reference key, evaluator instructions, D1 and D2, gold annotation and metrics are in the public repository. All code was written during the sprint; the README marks the pre-existing materials (D0 data, ICML coding instrument

and Q1 prompt). The English protocol description in protocol/ was added after the freeze (tag protocol-v2.1) and says so. The synthetic user profiles contain no personal data. Potential dual-use risks and safeguards are discussed in Appendix A.

Code repository: https://github.com/obohatov/apart-epistemics-sprint (tags protocol-v2.1 to d2-v1.0)
Data: data/D1, data/D2, annotation/ and data/eval/ in the same repository.

Author Contributions (optional)

O.B. designed the study, wrote the reference key, annotated the gold, approved the perturbations, performed the blind matching and wrote and verified the report. O.B. is the founder of VeriHub, a company that audits how public LLMs represent organisations; this is a potential conflict of interest in the choice of topic. The study reused the author's pre-existing ICML research dataset (D0) for evaluator development.

References

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Appendix (optional)

Appendix A. Limitations and Dual-Use / Ethical Considerations

What the results establish. On three Brussels questions, 18 gold answers and six perturbation bases, evaluator C did not detect more issues than evaluator B when both used the same model and reference pack. They do not establish that B and C are equivalent in general, that the rubric is useless, or that either evaluator can certify an answer. The gold has 33 issues, so intervals are wide.

Methodological limitations. (i) One annotator, who also wrote the reference key: gold and evaluator pack share a source, which favours evaluators on issues the key covers and hides issues it does not cover. (ii) The 11 issues missed by both may reflect key gaps; if so, recall is bounded by the key. (iii) D2 uses six bases and author-written perturbations; expected severity was set by the author. (iv) The base model of the evaluators and one generator are from the same family; evaluators flag Claude answers more often (B 21/27, C 22/27 vs GPT 13/27, 12/27), which may reflect real quality differences or bias, and our gold sample cannot separate them. (v) B and C differ in several ways at once. (vi) Answers are API answers with web search on 13 Nov 2026 and the rules change; D0 answers came from consumer interfaces. (vii) Evaluators were tuned on D0 only and we did not test prompt or seed variation. (viii) Severity agreement is computed on matched pairs only.

Dual use. Detectors of this kind can be misused: (a) to tune answers until they pass the evaluator without becoming correct, since the evaluator only sees what the key covers; (b) to give false assurance that an unflagged answer is complete; (c) to attack an organisation or service by publishing selected flags about how assistants describe it, without the noise level (44% of flags were false alarms). We recommend reporting flag precision and recall next to any result, using evaluators for triage and not certification, and keeping a human reviewer for material cases.

Personal data. No personal data were collected. User profiles are synthetic, no human subjects were involved, and the annotation was done by the author on model outputs. Conflict of interest: see Author Contributions.

Appendix B. Pre-existing versus sprint work

Component	Status
ICML 2026 study, D0 (18 chat responses), interpretive-fidelity rubric and coding instrument, Q1 prompt	Pre-existing
Protocol v2.1, prompts Q2–Q3, reference key and authority map, FR/NL translations	Sprint
D1 collection code and data (54 answers), blind gold (18 answers, 33 issues)	Sprint
D2 cards and objects (36), evaluators B and C, evaluation and metrics code	Sprint
Blind matching (122 D1 flags, 230 D2 flags), final metrics, this report	Sprint

Appendix C. Gold issues missed by both evaluators

GI07, GI10, GI11, GI12, GI13, GI19, GI20, GI24, GI29, GI30, GI31 (11 of 33). Most concern subtle conditional rules. Only B found GI04 and GI22; only C found GI14 and GI28. Full descriptions are in annotation/blind/D1_gold_issues.csv.

LLM Usage Statement

Claude (Anthropic) was used throughout: to draft and review code, prompts and translations, to run the evaluation pipeline, to compute metrics and to draft this report; a second AI assistant independently reviewed the research protocol, evaluator instructions and D2 perturbation design and gave methodological critiques. The author retained responsibility for the reference key, gold annotation, perturbation approval, blind matching and all conclusions. Every number in this report was recomputed from the repository files and checked against analysis/final_metrics.md.